

Digital Forensics Laboratory

EXPERIMENT NO: 02

TITLE: Recover deleted or damaged files from a storage device using TestDisk

1. AIM

To recover deleted files and repair a missing or corrupted partition from a storage device using the TestDisk data recovery command-line utility.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** TestDisk (v7.3-WIP Data Recovery Utility)
- **Hardware:** Target Storage Device (e.g., hp v221w 16GB USB Device).

3. THEORY

TestDisk is a powerful, open-source data recovery software primarily designed to help recover lost partitions and make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses, or human error (such as accidentally deleting a Partition Table). It can also be utilized to undelete files from FAT, exFAT, NTFS, and ext2 file systems. TestDisk queries the BIOS or the OS in order to find the hard drives and their characteristics, performs a quick check of the disk structure, and looks for lost partitions or deleted files.

4. PROCEDURE

Step 1: Log Creation & Disk Selection

1. Open the testdisk_win.exe application as an Administrator.
2. **Log Creation:** Choose [Create] to instruct TestDisk to create a log file containing technical information and messages. Press Enter to proceed.

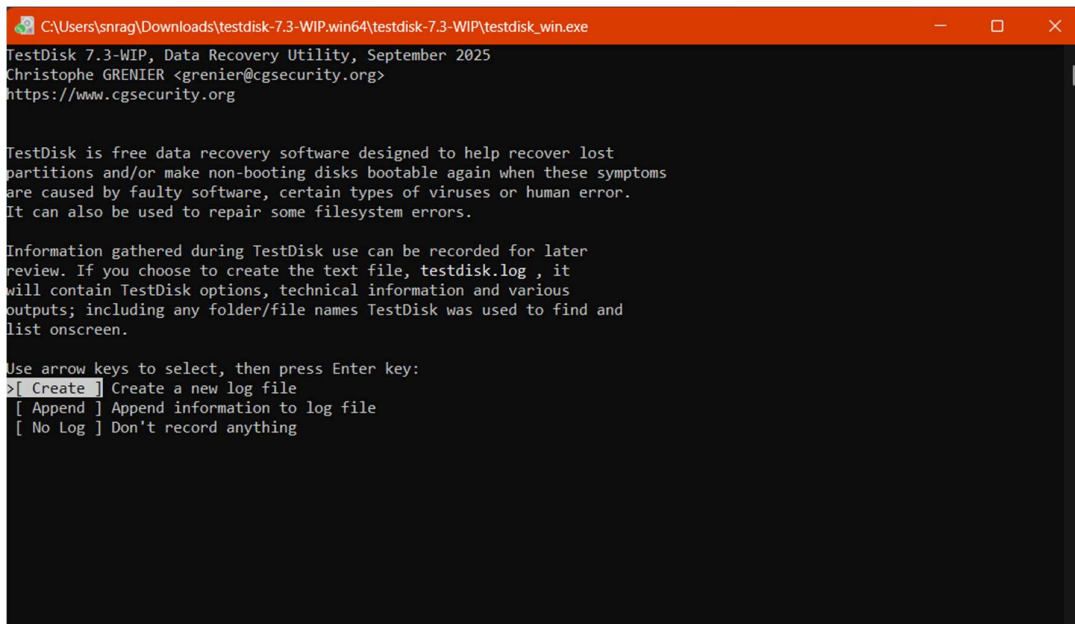


Figure 1: Initial TestDisk dashboard prompting for log creation

3. **Disk Selection:** All connected hard drives will be detected and listed. Use the up/down arrow keys to select the target hard drive containing the lost partitions or deleted files (e.g., Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w). Press Enter to Proceed.

```
C:\Users\snrag\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - NVMe Sandisk PC SN5000S SDEPNSJ-512G-1132
>Disk \\.\PhysicalDrive1 - 30 GB / 28 GiB - SanDisk Cruzer Blade

>[Proceed] [Quit]
```

Note: Serial number 03020302040924193359
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.

Figure 2: Selecting the physical storage drive in TestDisk

Step 2: Partition Table Type Selection

1. TestDisk displays the available partition table types.
2. Select the partition table type. Usually, the default value is the correct one as TestDisk auto-detects the partition table type (e.g., [Intel] Intel/PC partition). Press Enter to Proceed.

```
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Disk \\.\PhysicalDrive1 - 30 GB / 28 GiB - SanDisk Cruzer Blade

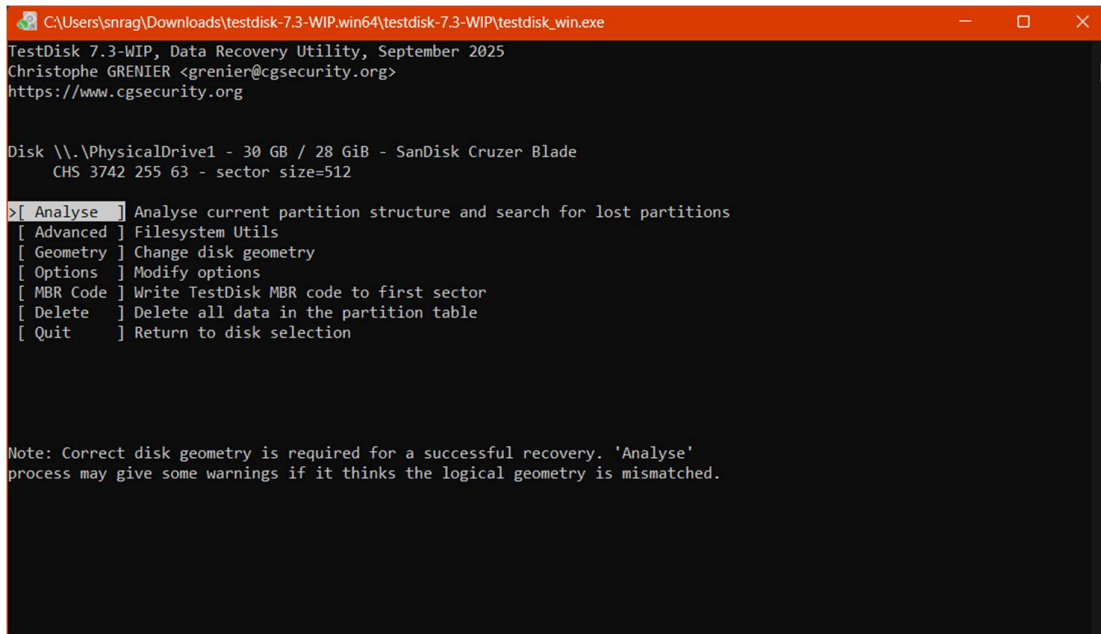
Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

Figure 3: Selecting the Intel/PC partition table type

Step 3: Current Partition Table Status

1. TestDisk displays the main menu options. Use the default menu [Analyse] to check your current partition structure and search for lost partitions. Confirm at Analyse with Enter.



```
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TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 30 GB / 28 GiB - SanDisk Cruzer Blade
CHS 3742 255 63 - sector size=512

>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

Figure 4: Choosing the Analyse option to examine the partition structure

2. The current partition structure is listed. Examine the structure for missing partitions and errors (e.g., Invalid NTFS boot, missing logical partitions). Confirm at [Quick Search] to proceed.

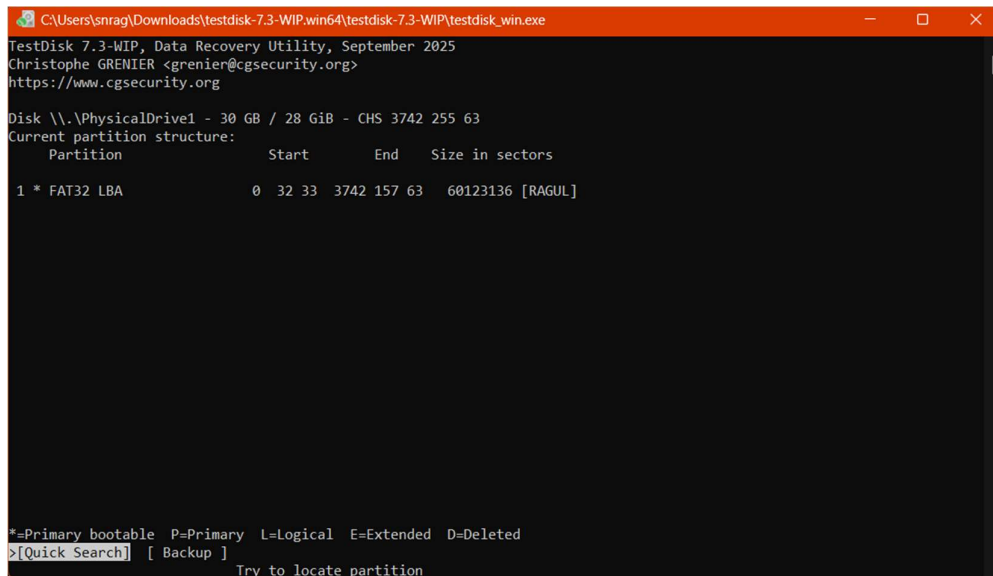


Figure 5: Initiating a Quick Search for lost partitions

Step 4: Deeper Search (If Required)

1. If the quick search does not find the required partition or if partitions overlap (status 'D' for deleted), highlight the [Deeper Search] menu and press Enter.
2. Deeper Search will scan each cylinder for FAT32 backup boot sectors, NTFS backup boot superblocks, etc., to detect more partitions.

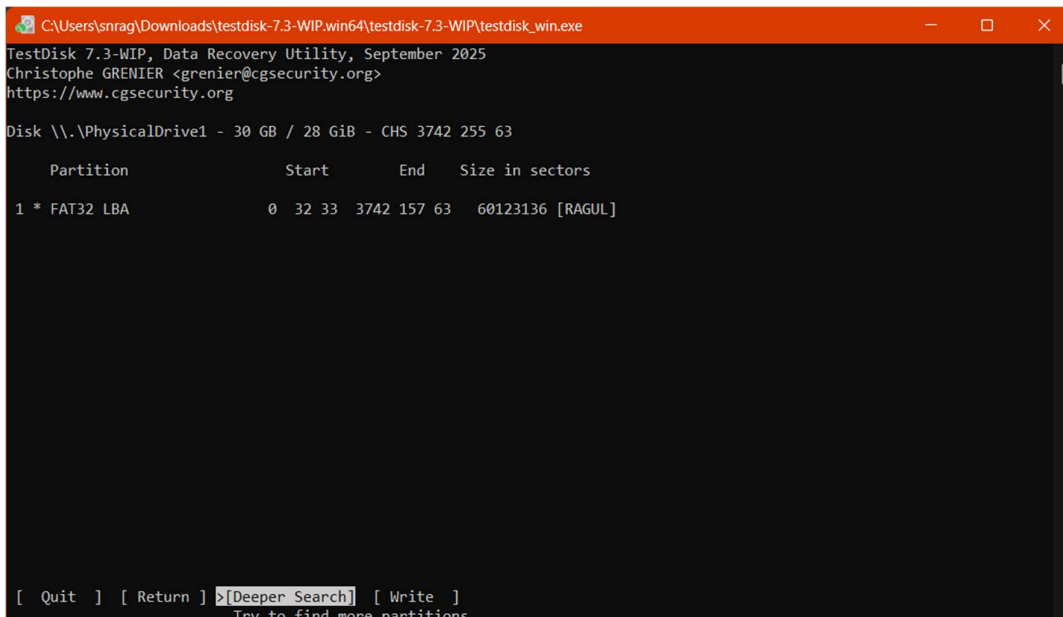


Figure 6: Selecting the Deeper Search option

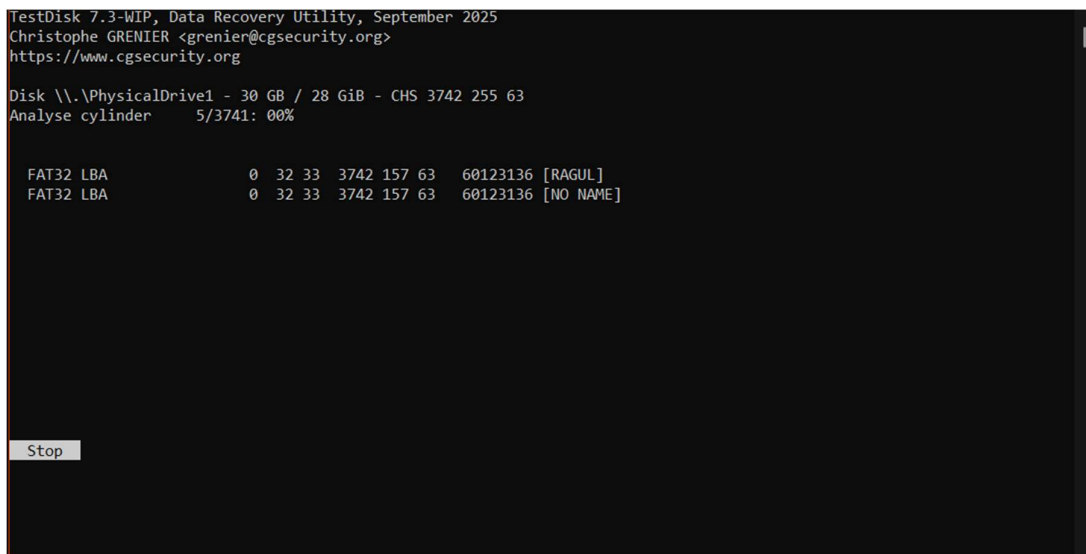


Figure 7: Deeper Search actively scanning disk cylinders

Step 5: Identifying and Accessing the Correct Partition

1. Once the search is complete, highlight the correct partition you wish to recover or extract files from (e.g., >L FAT32 LBA).
2. Press the p key to list the files inside that partition.

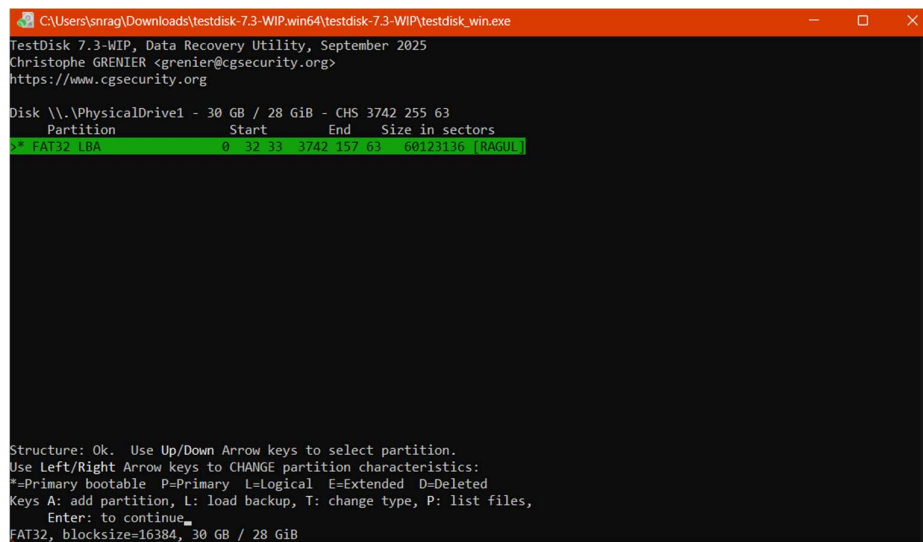
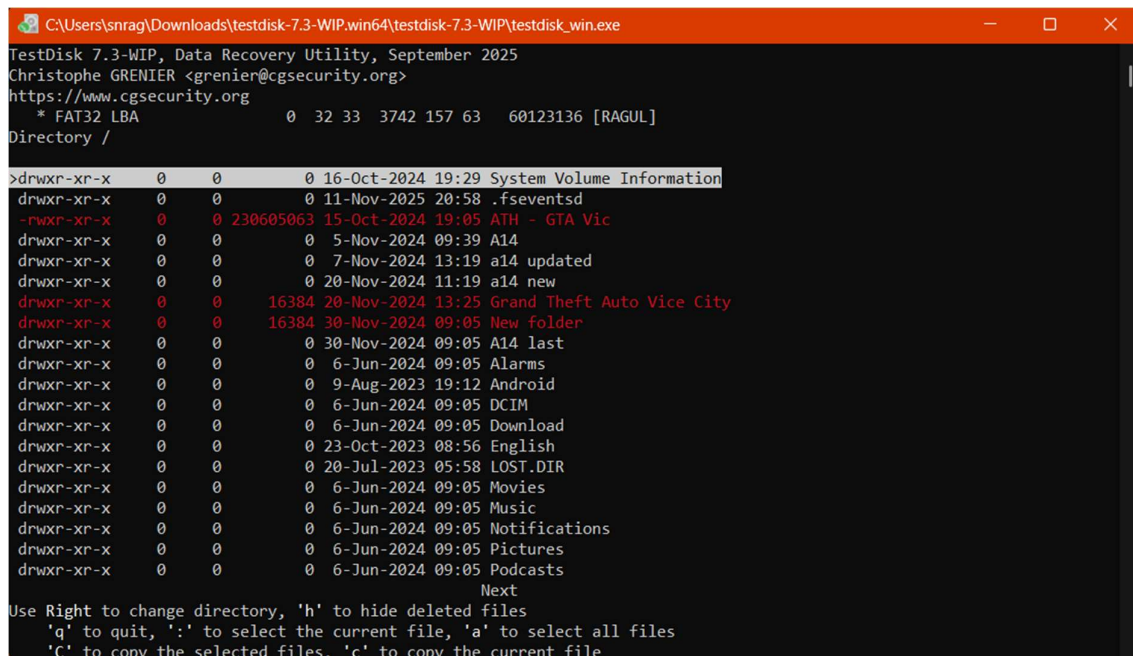


Figure 8: Highlighting the located FAT32 partition

Step 6: Recovering / Copying Deleted Files

1. In the file listing, navigate through the folders using arrow keys. Files listed in red indicate deleted entries.
2. Highlight the specific deleted file you want to recover (e.g., images (1).jpg).



```
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TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32 LBA 0 32 33 3742 157 63 60123136 [RAGUL]
Directory /
>drwxr-xr-x 0 0 0 16-Oct-2024 19:29 System Volume Information
drwxr-xr-x 0 0 0 11-Nov-2025 20:58 .fsevents
-rwxr-xr-x 0 0 230605063 15-Oct-2024 19:05 ATH - GTA Vic
drwxr-xr-x 0 0 0 5-Nov-2024 09:39 A14
drwxr-xr-x 0 0 0 7-Nov-2024 13:19 a14 updated
drwxr-xr-x 0 0 0 20-Nov-2024 11:19 a14 new
drwxr-xr-x 0 0 16384 20-Nov-2024 13:25 Grand Theft Auto Vice City
drwxr-xr-x 0 0 16384 30-Nov-2024 09:05 New folder
drwxr-xr-x 0 0 0 30-Nov-2024 09:05 A14 last
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Alarms
drwxr-xr-x 0 0 0 9-Aug-2023 19:12 Android
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 DCIM
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Download
drwxr-xr-x 0 0 0 23-Oct-2023 08:56 English
drwxr-xr-x 0 0 0 20-Jul-2023 05:58 LOST.DIR
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Movies
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Music
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Notifications
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Pictures
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Podcasts
Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

Figure 9: Listing the directory contents and selecting a file for recovery

3. Press c to copy the current selected file (or C to copy all selected files).
4. You will be prompted to select a destination directory to save the recovered file. Navigate to the desired local folder and press C when the destination is correct.
5. TestDisk will display a confirmation message indicating a successful recovery (e.g., "Copy done! 1 ok, 0 failed").

```
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* FAT32 LBA 0 32 33 3742 157 63 60123136 [RAGUL]
Directory /Download
Copy done! 0 ok, 0 failed
drwxr-xr-x 0 0 0 16-Oct-2024 19:29 System Volume Information
drwxr-xr-x 0 0 0 11-Nov-2025 20:58 .fsevents
-rwxr-xr-x 0 0 230605063 15-Oct-2024 19:05 ATH - GTA Vic
drwxr-xr-x 0 0 0 5-Nov-2024 09:39 A14
drwxr-xr-x 0 0 0 7-Nov-2024 13:19 a14 updated
drwxr-xr-x 0 0 0 20-Nov-2024 11:19 a14 new
drwxr-xr-x 0 0 16384 20-Nov-2024 13:25 Grand Theft Auto Vice City
drwxr-xr-x 0 0 16384 30-Nov-2024 09:05 New folder
drwxr-xr-x 0 0 0 30-Nov-2024 09:05 A14 last
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Alarms
drwxr-xr-x 0 0 0 9-Aug-2023 19:12 Android
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 DCIM
>drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Download
drwxr-xr-x 0 0 0 23-Oct-2023 08:56 English
drwxr-xr-x 0 0 0 20-Jul-2023 05:58 LOST.DIR
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Movies
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Music
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Notifications
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Pictures
drwxr-xr-x 0 0 0 6-Jun-2024 09:05 Podcasts
Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'c' to copy the selected files, 'C' to copy the current file
```

Figure 11: Successful copy/recovery of the deleted file

Step 7: Partition Table and Boot Sector Recovery (Optional)

1. Press q to quit the file listing and return to the partition selection.
2. Use the left/right arrow keys to change the status of a partition from D (Deleted) to L (Logical) or P (Primary) to recover the partition itself. Press Enter.
3. If all partitions are correctly listed, confirm at [Write] with Enter, press y to confirm, and then OK.
4. If a boot sector is damaged, use the [Backup BS] option to copy the backup boot sector over the damaged one.
5. Exit TestDisk and reboot the computer to access the restored partition data.

5. RESULT

The TestDisk data recovery utility was successfully executed to analyze the physical drive's geometry and partition tables. Lost partitions were successfully identified through deeper searching, and deleted files were effectively navigated, selected, and copied (recovered) to a secure local destination.